

4.4.1 Plant reproduction

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the control of flowering in plants	To include the role of vernalisation and phytochrome in the timing of flowering. HSW2, HSW5, HSW8
(b) adaptations of flowers for pollination	To include adaptations of both wind-pollinated flowers (e.g. cereals) and insect-pollinated flowers (e.g. legumes). PAG2 HSW4, HSW5, HSW8
(c) fertilisation and seed formation	To include the growth of the pollen tube, the formation of the embryo and the formation of the endosperm (in outline only). PAG2 HSW4, HSW5, HSW8
(d) (i) the germination of seeds (ii) investigations into the factors affecting germination	To include the role of gibberellin as a cell signalling molecule. HSW1, HSW2, HSW8
(e) the importance of cereals as staple foods	To include a consideration of the global use of rice, maize and wheat, and issues relating to food security and sustainability. HSW9, HSW10, HSW12